BIOGRAPHICAL SKETCH

NAME: Mathews, Rick

POSITION TITLE: MD Candidate

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	END DATE MM/YYYY	FIELD OF STUDY
William E. Macaulay Honors College at the City University of New York at the City College of New York, New York, NY	BENG	06/2017	Biomedical Engineering
Oregon Health & Science University, Portland, OR	PHD	12/2024	Biomedical Engineering
Oregon Health & Science University, Portland, OR	MD	2027	Medicine

A. Personal Statement

I am an MD/PhD candidate at the Oregon Health & Science University (OHSU) in Portland, Oregon with a background in biomedical engineering. My research focuses on vascular pathophysiology. My current focus is on a congenital cardiovascular lesion known as coarctation of the aorta. This is a relatively unexplored lesion due to the paucity of pathologic samples and difficulty in obtaining control tissue to conduct rigorous scientific investigation. Systematic efforts at the Doernbecher Children's Hospital in coordination with the pediatric cardiac surgery team has established an IRB approved and patient consented tissue and data repository enabling this work. The *Friends of Doernbecher Foundation* has provided funding to map a missing baseline for this disease, a framework we outline in our publication in the *American Journal of Physiology* in 2026. Further review demonstrated robust expression of a prenatal marker in coarctation resections, *TFAP2B*, when published mammalian data investigating this marker at the relevant anatomic junction refutes its postnatal persistence; we have released a preprint on *Research Square* detailing this finding in a clinical case.

1. **Mathews R**, Hinds MT, Woltjer RL, Muralidaran A. Territory mapping the ductal-aortic junction throughout gestation is a critical and missing baseline for coarctation of the aorta. Am J Physiol Heart Circ Physiol. 2026 Mar 1;330(3):H869-H873. doi: 10.1152/ajpheart.00741.2025. Epub 2026 Feb 12. PMID: 41677334.

2. **Mathews R,** Ditmore R, Krajbich V, Hinds MT, Woltjer RL, Muralidaran A. Heterochronic Expression of TFAP2β in Coarctation of the Aorta. Res Sq [Preprint]. 2026 Jun 30:rs.3.rs-10021035. doi: 10.21203/rs.3.rs-10021035/v1. PMID: 42427849; PMCID: PMC13345516.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2021 - 2022	Platform and Portfolio Company Support, Groove Capital, Minneapolis, MN
2020 - 2024	PhD Candidate, Oregon Health & Science University, Portland, OR
2018 -	MD Candidate, Oregon Health & Science University, Portland, OR
2018 - 2021	Director, Board of Trustees, Phi Beta Pi: Marquam Hill Cooperative 501(c)(3), Portland, OR
2018 - 2018	Project Manager, Pathology Applications, Epic Systems, Verona, WI
2016 - 2018	Co-founder, Gol: Water Filtration Solutions for Developing Areas, New York, NY
2016 - 2017	Emergency Room Volunteer, The New York-Presbyterian Brooklyn Methodist Hospital, Brooklyn, NY
2015 - 2017	Member, The College Group at the Metropolitan Museum of Art, New York, NY
2015 - 2015	Educational Testing Consultant, The Grand Street Settlement 501(c)(3), New York, NY
2014 - 2018	Research Assistant, Wallace H. Coulter Laboratory for Cardiovascular Dynamics and Biomolecular Transport, New York, NY
2014 - 2017	Member, The National Engineering Honor Society - Tau Beta Pi, New York, NY
2014 - 2015	Head Teaching Assistant, The Chemistry Department at the City University of New York at the City College of New York, New York, NY

Honors

2026 - 2026	Alpha Omega Alpha Honor Medical Society
2026 - 2027	Friends of Doernbecher Grant, OHSU Foundation
2022 - 2026	Ruth L. Kirschstein Individual Predoctoral NRSA for MD/PhD and other Dual Degree Fellowships, The National Heart, Lung, and Blood Institute. Grant F30HL163918
2026 - 2026	Robert Bigley Scholarship
2023 - 2024	N.L. Tartar Trust Award, OHSU School of Medicine and Department of Biomedical Engineering
2022 - 2022	Student Travel Award, The Pacific Northwest Vascular Society
2022 - 2022	Looking to the Future Scholar, The Society of Thoracic Surgeons
2022 - 2022	Vascular Annual Meeting Travel Award, The Society for Vascular Surgery
2020 - 2022	Multidisciplinary Research Training in Pulmonary and Critical Care Medicine, The National Heart, Lung, and Blood Institute. Grant T32HL083808
2021 - 2021	Walter de Groot Scholar, The American Vein and Lymphatic Society Annual Meeting
2018 - 2020	Medical Scientist Training Program, The National Institute of General Medical Sciences. Grant T32GM109835
2017 - 2017	Wallace H. Coulter Award for Outstanding Undergraduate Research, The Biomedical Engineering Department at the City University of New York at the City College of New York
2017 - 2017	Seymour A. Seiler Undergraduate Research Fund Award, William E. Macaulay Honors College at the City University of New York at the City College of New York
2015 - 2015	Grant for Undergraduate Research in Biomedical Engineering, The New York Community Trust
2013 - 2017	Full Tuition Scholarship, William E. Macaulay Honors College at the City University of New York at the City College of New York
2013 - 2017	Dean's List, William E. Macaulay Honors College at the City University of New York at the City College of New York

C. Contribution to Science

1. Coarctation of the aorta (CoA) is a lesion with a surgical solution and no definitive etiologic explanation. Presenting on a spectrum between simple and complex CoA, the common denominator is a discrete, obstructive shelf of tissue protruding into the aortic lumen at the junction of the ductus arteriosus and the aorta. Major unanswered questions remain, including: is there a common developmental origin for this obstructive shelf of tissue and if so, when and how did it deviate from normal development? Multiple theories attempt to explain the pathology of CoA; however, none of them reference normal fetal development of the ductal-aortic junction. We identify a critical and missing baseline that is addressable today and would help bridge theory to mechanistic investigation.
   a. Mathews R, Hinds MT, Woltjer RL, Muralidaran A. Territory mapping the ductal-aortic junction throughout gestation is a critical and missing baseline for coarctation of the aorta. Am J Physiol Heart Circ Physiol. 2026;330(3):H869-H873. doi:10.1152/ajpheart.00741.2025
   b. Mathews R, Ditmore R, Krajbich V, Hinds MT, Woltjer RL, Muralidaran A. Heterochronic Expression of TFAP2β in Coarctation of the Aorta. Res Sq [Preprint]. 2026 Jun 30:rs.3.rs-10021035. doi: 10.21203/rs.3.rs-10021035/v1. PMID: 42427849; PMCID: PMC13345516.

2. Venous thrombosis levies significant morbidity and mortality on the global population. Significant barriers remain in the diagnosis, management, and treatment of this lesion in various settings (Mathews et. al, Current Opinion Hematology 2024). Through a mouse model of deep vein thrombosis (Mathews et. al, Platelets 2024) we modeled the role of platelet count in determining the burden of deep vein thrombi in mice and associated a higher thrombus burden with higher platelet counts. In the setting of veno-venous ECMO, we presented data demonstrating better outcomes associated with direct thrombin inhibition using Bivalirudin vs Heparin (Kartika et al, Eur J Haematol 2024).
   a. Mathews R, Setthavongsack N, Le-Cook A, Kaempf A, Loftis JM, Woltjer RL, Lorentz CU, Revenko A, Hinds MT, Nguyen KP. Role of platelet count in a murine stasis model of deep vein thrombosis. Platelets. 2024 Dec;35(1):2290916. PubMed Central PMCID: PMC10805383.

b. Mathews R, Hinds MT, Nguyen KP. Venous thromboembolism: diagnostic advances and unaddressed challenges in management. Curr Opin Hematol. 2024 May 1;31(3):122-129. PubMed Central PMCID: PMC10977858.

c. Kartika T, Mathews R, Migneco G, Bundy T, Kaempf AJ, Pfeffer M, DeLoughery TG, Moore K, Beardshear R, Oetken HJ, Case J, Hinds MT, McCarty OJT, Shatzel JJ, Zonies D, Zakhary B. Comparison of bleeding and thrombotic outcomes in veno-venous extracorporeal membrane oxygenation: Heparin versus bivalirudin. Eur J Haematol. 2024 Apr;112(4):566-576. PubMed Central PMCID: PMC11034845.

3. The endothelial glycocalyx (eGCX), a membrane-bound layer of proteoglycans, glycoproteins, and glycosaminoglycans, has emerged as a critical regulator of vascular physiology. Beyond its role as a physical barrier, the eGCX influences key processes including immunomodulation, cell proliferation, surface receptor activation, and vasoregulation. Our research investigates eGCX-mediated mechanisms underlying endothelial cell response to physiologic and pathologic mechanical stimuli. Specifically, we previously demonstrated the coordinated roles of Glypican-1 and PECAM-1 in mediating shear-induced nitric oxide (NO) production (Bartosch et al, Biophys J 2017, Sci Rep 2021) and examined the response of the eGCX to shear under inflammatory conditions (Mathews et al, BBRC 2024). Further exploring the influence of the vascular microenvironment, we investigated how age-related stiffening of the basement membrane alters endothelial cell phenotype. Our findings implicate Glypican-1 as a key protector against the detrimental effects of increased basement membrane stiffness, mitigating inflammation, dysregulated NO production, and endothelial-mesenchymal transition (EndoMT) (Mahmoud et al, Cardiovasc Res 2021).
   a. Mathews R, Pang J, Muralidaran SA, King CGX, McCarty OJT, Hinds MT. AMP dependent protein kinase regulates endothelial heparan sulfate expression in response to an inflammatory stimulus under arterial shear stress. Biochem Biophys Res Commun. 2024 Nov 26;735:150743. PubMed Central PMCID: PMC11601061.

   b. Bartosch AMW, Mathews R, Mahmoud MM, Cancel LM, Haq ZS, Tarbell JM. Heparan sulfate proteoglycan glypican-1 and PECAM-1 cooperate in shear-induced endothelial nitric oxide production. Sci Rep. 2021 May 31;11(1):11386. PubMed Central PMCID: PMC8166914.

   c. Mahmoud M, Mayer M, Cancel LM, Bartosch AM, Mathews R, Tarbell JM. The glycocalyx core protein Glypican 1 protects vessel wall endothelial cells from stiffness-mediated dysfunction and disease. Cardiovasc Res. 2021 May 25;117(6):1592-1605. PubMed Central PMCID: PMC8152694.

   d. Bartosch AMW, Mathews R, Tarbell JM. Endothelial Glycocalyx-Mediated Nitric Oxide Production in Response to Selective AFM Pulling. Biophys J. 2017 Jul 11;113(1):101-108. PubMed Central PMCID: PMC5510764.

Complete List of Published Work in My Bibliography:
https://www.ncbi.nlm.nih.gov/myncbi/rick.mathews.1/bibliography/public/